

Antiatherosclerotic effects of dietary supplementations of garlic and turmeric: Restoration of endothelial function in rats

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Abstract

Protections of endothelial integrity by elimination of certain risk have proven to be effective in maintaining hemostasis and in slowing the progress of the cardiovascular disease. Indigenous drugs are the natural source of protection against these disorders, which can be used more effectively by the knowledge of their active ingredients as well as by their mechanism of action. Most prominent among these drugs are garlic, [*Alium sativum* L., Family: Liliaceae, Bulbs] and turmeric [*Curcuma longa* L., Family: Zingiberaceae, Rhizomes]; commonly used Indian traditional spices. In the present study, we examined the atheroscleroprotective potential of diet supplementation of garlic and turmeric by measuring serum lipid profile, changes in cardiovascular parameters i.e. arterial blood pressure, electrocardiogram and heart rate. We further tried to elucidate the mechanism of restoration of endothelial function and the role of endothelium-derived factors mainly, nitric oxide (NO) and cyclooxygenase derived contracting factors. A notable restoration of arterial blood pressure was seen in animals on garlic and turmeric supplemented diet. Animals on supplemented diet showed a significantly enhanced vasorelaxant response to adenosine, acetylcholine, isoproterenol and contractile effect of 5-hydroxytryptamine was significantly attenuated. Inhibition of these responses by L-NMMA was smaller in tissues from herbal treated animals. Incubation of tissues with L-arginine (10^{-5} M) resulted in a significant reversal of L-NMMA induced inhibition of endothelium-mediated relaxation, which appeared to be pronounced in rings from animals supplemented with herbs as compared to hypercholesterolemic animals. Addition of indomethacin (10^{-5} M) augmented the relaxation in all the groups of animals. The present study demonstrated that garlic and turmeric are potent vasorelaxants as well as reduce

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the atherogenic properties of cholesterol. Whether combination of these vasodilators in cardiovascular disorders with increased peripheral vascular resistance remains to be determined.

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Introduction

Endothelium, a single layered mechanical barrier between the circulating blood and the media and adventitia of the blood vessels, constitutes one of the largest and important organs. It regulates the various important functions of cardiovascular system such as vascular tone, regulation of blood flow, proliferation of smooth muscle etc. Endothelium releases variety of substances like prostacyclins, growth factors, endothelin-1, angiotensin II, other vasoconstricting substances, hyperpolarizing factors and most importantly nitric oxide (NO): the main endothelium derived relaxing factor (EDRF) (Rubanyi, 1993; Landmesser et al., 2004; Vanhoutte and Boulanger, 1995; Kitamoto and Egashira, 2004).

Abnormalities in endothelium-dependent vasomotion are detectable in many chronic vascular disease states, such as hypertension, diabetes and atherosclerosis. These abnormalities in smooth muscle cells (SMCs) reactivity are thought to be the result of both decreased endothelial NO production and increased destruction of labile NO locally within diseased vasculature (Biegelsen and Loscalzo, 1999). The decrease in NO availability is attributed to superoxide production (Stokes et al., 2002) and oxidative modifications of lipoproteins (Fan and Watanabe, 2003) resulting in deleterious effects on endothelium in diseased conditions.

The protection of endothelial integrity by elimination of certain risk factors via lipid lowering agents (Rosenson, 2001), free radical scavengers and antioxidants (Antoniades et al., 2003) have proven to be effective in restoring endothelial function and in slowing the progress of the disease (Rubanyi, 1993). Ironically most of the synthetic drugs in use are in their pure and active forms and many of these compounds are known to have adverse effects on heart and other organs of the body.

Those looking for an alternative therapy have tried natural herbs. Certain indigenous drug preparations with naturally occurring herbs have been in use for many decades for the treatment of cardiovascular diseases (Thompson Coon and Ernst, 2003; Ghasi et al., 2000; Kris-Etherton et al., 2002). However, the underlying mechanism of action of these agents is not clearly understood. Most prominent among these herbs are commonly used Indian traditional spices: garlic, [*Alium sativum* L., Family: Liliaceae, Bulbs] and turmeric [*Curcuma longa* L., Family: Zingiberaceae, Rhizomes].

There are several reports supporting the pharmacological importance of garlic for hypoglycemic, hypolipidemic, antimicrobial, antitumor, antioxidant and antiatherogenic (preventive) as well as antiatherosclerotic effects (Elkayam et al., 2003; Alder et al., 2003; Banerjee and Maulik, 2002; Brace, 2002; Agarwal, 1996; Orekhov et al., 1995). It is believed that garlic inhibits blood coagulation, platelet aggregation and also causes vasodilation (Aqel et al., 1991; Harenberg et al., 1988). The organo-sulfur compounds e.g. allicin and diallyl disulfide are considered to be responsible for most of the pharmacological activity of garlic (Agarwal, 1996). In addition, other alliin (an odorless precursor of allicin) derived compounds such as ajoene (a nonvolatile compound) and several amino acids especially arginine, are present in garlic (Agarwal, 1996). Recent studies

have demonstrated that garlic exerts its therapeutic effect by increasing NO production (Maslin et al., 1997). It is also found to have free radical scavenging action and inhibits oxidative modification of low-density lipoproteins. Earlier studies including our own (Ashraf et al., 1999) have shown that garlic has a relaxant effect on vascular smooth muscle, which appears to be mediated through endothelium (Aqel et al., 1991).

Turmeric (*C. longa*) is a perennial herb with large rootstocks consisting of rhizome. Curcumin (diferuloylmethane), a major component of turmeric and its analogs have a variety of physiological and pharmacological activities such as anti-inflammatory, anticarcinogenic and antioxidant properties (Azuine and Bhide, 1992). Turmeric provides protection against oxidative stress, and it has also been found to decrease lipid peroxidation in humans as well (Ramirez-Tortosa et al., 1999). Turmeric extract is reported to lower plasma cholesterol and triglycerides in hyperlipidemic rabbits (Ramirez-Tortosa et al., 1999; Deters et al., 2003).

In the present study, we examined the atheroscleroprotective potential of diet supplementation of garlic and turmeric by measuring serum lipid profile changes in cardiovascular parameters (arterial blood pressure, electrocardiogram and heart rate). These changes were monitored to examine the beneficial effects of herbal supplementation on overall physiology of animals. We further tried to elucidate the restoration of endothelial function with some known endothelial agonists and the role of endothelium-derived factors mainly, EDRFs; NO and cyclooxygenase derived contracting factors. The involvement of these mechanisms was pharmacologically evaluated using specific blockers in isolated tissue preparations.

Methods

Animals

All animal experiments were performed within the institutional guidelines. Adult healthy albino wistar rats of either sex, weighing 200 ± 25 g were used for the study. Rats were housed identically under a 12hr light: dark cycle. All the experimental animals were provided water and feed ad libitum.

Preparation of Garlic and Turmeric Extracts

Alium sativum L. (Family: Liliaceae) (Bulbs) and *Curcuma longa* L. (Family: Zingiberaceae) (Rhizomes) extracts were provided by Hamdard Laboratories, Delhi, India. Garlic cloves and turmeric rhizomes were washed extensively with sterilized water, surface fractions were removed aseptically, homogenized, squeezed out through double cheese cloth to obtain the extract and stored at -20°C until use.

Diet and Treatment

Animals were maintained on a standard diet (normal control) [%; crude protein—21.0, fat (ether extract)—5.0, crude fiber—4.0, ash—8.0, calcium—1.0, phosphorus—0.6, carbohydrate (nitrogen free extract)—53.0, metabolized energy (kcal/kg)—3600] (Hindustan Liver Ltd, India). In order to examine the endothelial dysfunction during experimentally induced atherosclerosis and therapeutic as well as

protective properties of herbal combination; a systematic study was performed on animals divided in the following groups.

Group I: Control animals on standard diet for 10 weeks.

Group II: Animals on (1% w/w) cholesterol supplemented feed for 10 weeks.

Group III: Animals on normal diet supplemented with (1% w/w) [garlic + turmeric, each 0.5% w/w] for 10 week.

Group IV: Animals on high cholesterol plus garlic + turmeric supplemented diet for 10 weeks.

Group V: Animals on cholesterol supplemented diet for 10 weeks, followed by cholesterol plus garlic + turmeric supplemented diet for 10 weeks.

In crushed feed cholesterol (Glaxo, India), garlic and turmeric powders were mixed for respective groups. The feed pellets were made with sterile water and dried in oven to avoid any contamination.

Measurement of Body Weight, Daily Food Intake and Other Biochemical Parameters

The body weight of each rat was determined before start of experiment and every week till the final experiment. Daily dietary food intake was monitored in animals from all groups. Blood samples were collected from overnight fasted animals. Serum cholesterol, triglycerides, high density lipoprotein (HDL) cholesterol, low density lipoprotein (LDL) cholesterol and very low density lipoprotein (VLDL) cholesterol were measured using enzymatic method employing commercial kit (Monozymes, India).

Arterial Blood Pressure and Heart Rate

Arterial blood pressure (ABP) and heart rate were recoded by earlier method (Fahim and Arndt, 1990). Briefly, animals were anaesthetized with α -chloralose (80mg/kg body weight) (BDH, UK). Later small doses of anaesthetic were administered intravenously whenever required. Trachea was cannulated for obstruction free breathing and body temperature of animal was maintained at 37–38°C. Femoral vein was cannulated for injections. Femoral artery on one side was exposed and heparinized, polyethylene cannula was inserted in the artery through small incision for recording ABP. The cannula was attached to a 23 gauge needle and connected via a three-way stopcock to pressure transducer (Statham, P23 Db). The pressure transducer was connected to cathode rays oscilloscope (TDS 420A, Tektronix, USA). Arterial blood pressure was recorded after 20 min of stabilization period after the surgery through a transbridge amplifier (TBM4, WPI, USA). Electrocardiogram (ECG) was obtained with silver bipolar electrodes (Lead II). Heart rate was calculated from EGC records.

Isolated Blood Vessel Preparations

Rats were sacrificed with an overdose of pentobarbitone sodium (80 mg/kg). Isometric tension of thoracic aorta was measured as described earlier (Fahim et al., 2001). Briefly, thoracic aorta was dissected, cut into small ring segments and were mounted in 10 ml organ bath containing Kreb's buffer

solution ([mM] NaCl-118.1, KCl-4.7, MgSO₄-1.2, KH₂PO₄-1.2, CaCl₂-5, NaHCO₃-5.0, Glucose-11.1) at $37 \pm 1^\circ\text{C}$ bubbled with carbogen gas (95% O₂ + 5% CO₂). Initial pre-load of 2 g was applied, after 1 hr for equilibration vessels were precontracted with 10^{-6} M phenylephrine (PE), after contraction reached plateau, agonists were added.

Experimental Design

The concentration-response relationships were determined by cumulative addition of acetylcholine (ACh) (10^{-13} – 10^{-4} M), adenosine (10^{-13} – 10^{-4} M), isoproterenol (IP) (10^{-13} – 10^{-4} M), 5 hydroxytryptamine (5-HT) (10^{-13} – 10^{-4} M) and sodium nitroprusside (SNP) (10^{-13} – 10^{-7} M) in aortic rings from all groups.

In order to determine the NO mediated effect of herbal supplements, the concentration-response curves for ACh, IP, adenosine and 5-HT were constructed before and 30 min after incubating the tissues with nitric oxide synthase (NOS) inhibitor L-NMMA (30 μM) in organ baths. The extent of NO mediated responses was further investigated by reversing the effects after incubating the tissue with NOS substrate L-arginine (10^{-5} M). The responses were also examined in the presence of glybenclamide (10^{-6} M) and indomethacin (10^{-5} M) to obtain the concentration-response curve without interference of other endothelial factors.

The concentration-response curve for ACh, IP, adenosine and 5-HT were obtained in the presence of glybenclamide (10^{-6} M) to eliminate the contribution of endothelium derived hyperpolarizing factors (EDHFs).

To examine the effect of herbal supplement on cyclooxygenase released contracting factors, concentration-response for ACh, IP, adenosine and 5-HT were obtained before and 30 min after treating the tissues with indomethacin (10^{-5} M). The response curves were obtained in the presence of glybenclamide (10^{-6} M) + L-NMMA (30 μM) to eliminate the contribution of other endothelial factors. One-half of maximally effective concentration (EC₅₀) value and maximum response (E_{max}) were determined from log concentration -response curve of each agonist.

Drugs

The following drugs were used; phenylephrine hydrochloride, indomethacin, glybenclamide, N^G-monomethyl-L-arginine (L-NMMA) (Sigma Chemical Co., USA). Phenylephrine HCl and L-NMMA were dissolved in distilled water. Indomethacin was dissolved in Na₂CO₃ (0.2 M) solution and glybenclamide in DMSO solution. Serial dilutions were made in triple distilled water.

Statistical Analysis

The steady state contraction produced by 10^{-6} M PE was expressed as an increase in tension (g) from the resting level. The relaxant/contractile responses were expressed as a decrease/increase (percent) in steady state contraction. EC₅₀ was estimated graphically as concentration causing 50% relaxation (Ashraf et al., 1999). Graph-pad Prism, software was used for statistical analysis. The p value for statistical significance was set at 0.05 levels. Student's t-test for paired and unpaired data and analysis of variance (ANOVA) followed by student-Neumen-Keul's test were used wherever appropriate.

Results

Body Weight, Daily Food Intake and Serum Lipid Profile

The body weight after the different dietary supplements and daily food consumption are shown in [Table 1](#). No significant difference was observed in body weight and daily food intake among the groups at the end of the treatment.

Lipid profile of serum total cholesterol, triglycerides, HDL, LDL and VLDL of animals from all the groups are summarized in [Table 2](#). Elevated serum cholesterol, triglycerides, LDL and VLDL was observed in animals on cholesterol supplemented animals (group II). Supplementation of diet with herbs (group III, IV and V) restored the normal lipid profile and a notable increase in serum HDL/LDL ratio was observed. Blood LDL level was significantly ($p < 0.05$) lower (33.5 ± 2.93 mg/dl) in group IV animals as compared to hypercholesterolemic animals (group II, 104 ± 15.48 mg/dl). A significant decrease in atherogenic index (total cholesterol /HDL cholesterol) in herbal supplemented animals, suggests the atheroprotective potential of these herbs. The index was 6.76 ± 1.4 in group IV as compared to 17.91 ± 1.31 in group II ($p < 0.05$).

Cardiovascular Responses

The cardiovascular effects of high cholesterol diet and herbal supplemented diet in rats are summarized in [Table 3](#). Animals on high cholesterol diet (group II) showed an elevated systolic blood pressure (127.8 ± 4.3 mmHg, $p < 0.05$) compared to standard control diet animals (group I) (99.25 ± 1.6 mmHg). Higher systolic pressure was also observed in group of animals with high cholesterol as well as herbal supplemented diet (group IV and V). Animals on herbal supplemented diet (group IV and V) showed a significantly ($p < 0.05$) lower diastolic pressure as compared to hypercholesterolemic animals (group II). Electrocardiogram (ECG) did not show any significant change in pattern in all groups of animals. However, a mild tachycardia was observed in hypercholesterolemic animals. No significant ($p > 0.05$) changes in heart rate were observed between the groups ([Table 3](#)).

Table 1

Effect of different diet supplements on final body weight and food intake in rats on normal feed (group I), cholesterol supplemented feed (group II), normal feed supplemented with turmeric + garlic (group III), cholesterol plus turmeric + garlic supplemented feed (group IV) and high cholesterol supplemented diet followed by turmeric + garlic plus cholesterol supplemented feed (group V)

Animal groups	I	II	III	IV	V
n	15	15	8	8	8
Body Weight (g)	255 ± 20	375 ± 50	250 ± 15	300 ± 25	300 ± 35
Food Intake (g/day)	18 ± 4	16 ± 2	20 ± 5	20 ± 5	17 ± 3

Data represent mean $\pm$ SEM, n= number of animals in each group.

Table 2

Effect of different diet supplements on blood lipid profile in rats on normal feed (group I), cholesterol supplemented feed (group II), normal feed supplemented with turmeric + garlic (group III), cholesterol plus turmeric + garlic supplemented feed (group IV) and high cholesterol supplemented diet followed by turmeric + garlic plus cholesterol supplemented feed (group V)

Animal groups	I	II	III	IV	V
n	15	15	8	8	8
Total Cholesterol (mg/dl)	80.25 ± 4.6	583 ± 26.7*	74.6 ± 8.8	179.2 ± 18 ⁺	290 ± 20.7 ⁺
Triglycerides (mg/dl)	85.6 ± 7.17	484.6 ± 63.27*	69.2 ± 8.8*	162.1 ± 5.5 ⁺	285 ± 20.7 ⁺
HDL (mg/dl)	47.67 ± 2.61	18.3 ± 1.63*	53.17 ± 2.27	35.5 ± 2.54 ⁺	28.5 ± 3.2 ⁺
LDL (mg/dl)	15.27 ± 2.6	104 ± 15.48	13.23 ± 1.45	33.5 ± 2.93 ⁺	64.7 ± 2.46 ⁺
VLDL (mg/dl)	6.43 ± 0.8	30.73 ± 6.59*	4.65 ± 1.44	14.27 ± 1.66 ⁺	24.3 ± 2.46

Data represent mean ± SEM, n = number of animals in each group.

*p < 0.05, group I vs group II or group III, ⁺p < 0.05 group II vs Group IV or group V.

Isolated aortic tissue response to acetylcholine, isoproterenol, adenosine and 5-hydroxytryptamine: effect of L-NMMA and L-arginine

Acetylcholine produced a concentration-dependent (10^{-13} – 10^{-4} M) relaxant response in all animals. However, there was attenuation to vasorelaxant response in cholesterol supplemented groups (group II, IV and V) (Fig. 1A, Table 4). Blood vessels from herbal supplemented animals showed a restoration of relaxation (groups IV and V) towards control (group I). The response was prominent in the concentration range of 10^{-8} to 10^{-4} M. The incubation of tissues with L-NMMA (30 μ M) caused a prominent ($p < 0.05$) reduction in relaxant response to ACh in all groups of animals (Fig. 1B). A significant rightward shift ($p < 0.05$) in concentration-response curve to group IV and group V vessels was observed as compared to hypercholesterolemic (group II) animals. However, the shift was significantly ($p < 0.05$) smaller in group IV and V, 29.74 ± 2.54 and $27.83 \pm 4.29\%$ respectively with 10^{-6} M ACh in comparison to group II ($19.75 \pm 2.31\%$) animals.

Incubation of tissues with L-arginine (10^{-5} M) resulted in a significant reversal of L-NMMA induced inhibition of ACh-induced relaxation (Fig. 1C, Table 4). In rings from animals supplemented with herbs (group IV and V) the reversal of ACh-induced relaxation was more pronounced as compared to hypercholesterolemic (group II) animals. In rings from group IV animals $54.98 \pm 3.51\%$ and in group V

Table 3

Effect of different diet supplements on cardiovascular parameters, in animals on normal feed (group I), cholesterol supplemented feed (group II), normal feed supplemented with turmeric + garlic (group III), cholesterol plus turmeric + garlic supplemented feed (group IV) and high cholesterol supplemented diet followed by turmeric + garlic plus cholesterol supplemented feed (group V)

Animal groups	I	II	III	IV	V
n	6	6	4	4	4
Systolic Pressure (mmHg)	99.25 ± 1.6	127.8 ± 4.3*	104.70 ± 2.9	110 ± 2.13 ⁺	115 ± 1.2 ⁺
Diastolic Pressure (mmHg)	64.0 ± 2.3	89.50 ± 3.59*	67.33 ± 1.66	72.5 ± 1.4 ⁺	77.25 ± 1.1
Mean Arterial Pressure (mmHg)	75.22 ± 1.67	104.3 ± 4.54*	79.75 ± 1.12	84.53 ± 1.15 ⁺	90.7 ± 1.12 ⁺
Heart Rate (b/min)	327.9 ± 3.1	307.3 ± 1.3	305.1 ± 1.75	312.7 ± 3.8	308.1 ± 1.34

Data represent mean ± SEM, n = number of animals in each group.

*p < 0.05 group I vs group II or group III, ⁺p < 0.05 group II vs Group IV or group V.

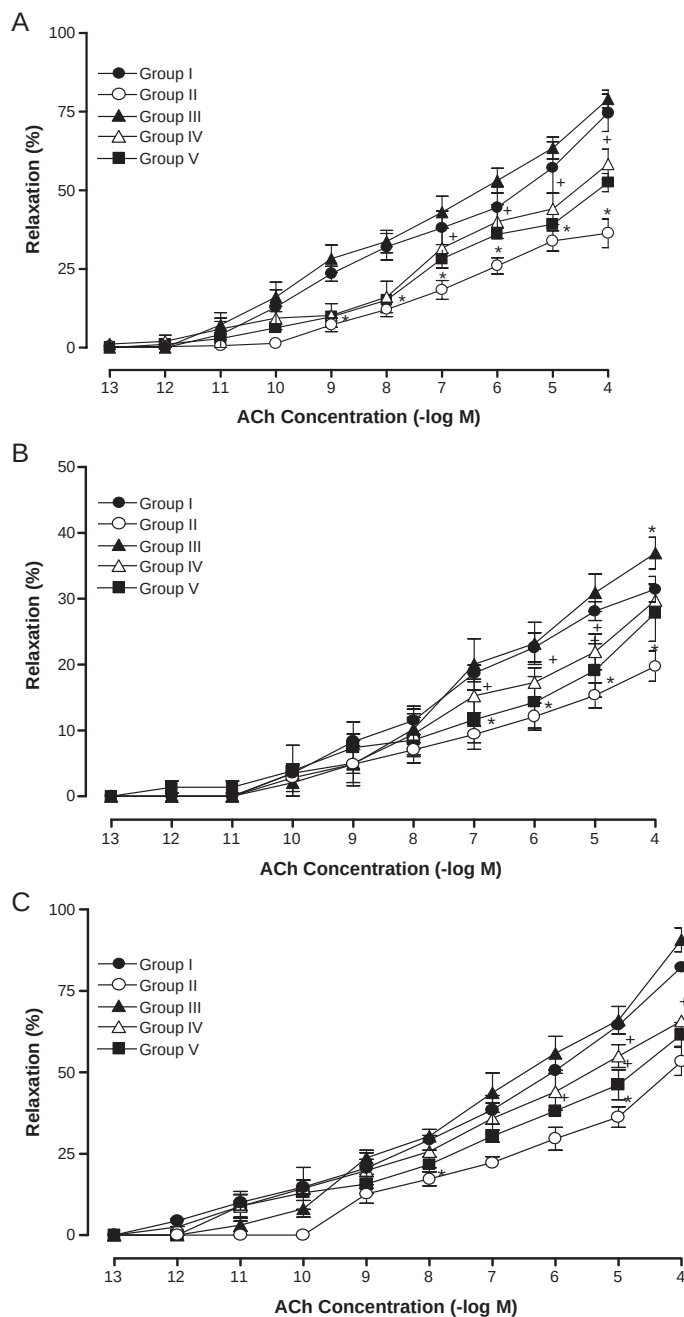


Fig. 1. Concentration-response (10^{-13} – 10^{-4} M) to acetylcholine (A), after incubation with 30 μ M L-NMMA (B) and 10^{-5} M L-arginine + 30 μ M L-NMMA (C) in 10^{-6} M PE precontracted intact-endothelium aortic-rings from rats on normal feed (group I), cholesterol supplemented feed (group II), normal feed supplemented with turmeric + garlic (group III), cholesterol plus turmeric + garlic supplemented feed (group IV) and high cholesterol supplemented diet followed by turmeric + garlic plus cholesterol supplemented feed (group V). Data represents mean $\pm$ SEM, 4 rings from each animal. * $p < 0.05$ group I vs group II or group III, + group II vs group IV or group V.

Table 4

Effective concentration of acetylcholine for 50% relaxation (EC_{50}) and maximum relaxant response—% change (E_{max}) to acetylcholine, before and after incubation of tissues with L-monomethylarginine (L-NMMA), L-NMMA + L-arginine, and indomethacin from rats on normal feed (group I), cholesterol supplemented feed (group II), normal feed supplemented with turmeric + garlic (group III), cholesterol plus turmeric + garlic supplemented feed (group IV) and high cholesterol supplemented diet followed by turmeric + garlic plus cholesterol supplemented feed (group V)

Groups	n	EC_{50} ($-\log M$)	E_{max}
<i>CONTROL</i>			
I	12	6.44 ± 0.28	88.8
II	10	—	44.5
III	10	$7.74 \pm 0.57^*$	92.5
IV	8	5.41 ± 0.39	65.3
V	8	5.81 ± 0.04	56.5
<i>L-NMMA (30 μM)</i>			
I	12	—	42.5
II	10	—	23.6
III	10	—	48.5
IV	8	—	38.85
V	8	—	32.55
<i>L-NMMA (30 μM) + L-ARGININE (10^{-5} M)</i>			
I	12	7.95 ± 0.32^a	95.5
II	10	$5.81 \pm 0.03^{*,a}$	60.23
III	10	7.51 ± 0.41	98.5
IV	8	$6.54 \pm 0.17^{+,a}$	73.41
V	8	$5.24 \pm 0.04^{+,a}$	69.77
<i>INDOMETHACIN (10^{-5} M)</i>			
I	10	7.55 ± 0.04	92.4
II	8	$6.63 \pm 0.09^{*,c}$	66.6
III	6	7.14 ± 0.09^c	96.36
IV	6	$6.10 \pm 0.05^{+,c}$	87.5
V	5	$6.16 \pm 0.07^{+,c}$	72.22

Data represent mean $\pm$ SEM, n = number of animals in each group, from each animal 4 ring were taken.

*— $p < 0.05$ group I vs group II or group III, ^a— $p < 0.05$ control vs L-NMMA or L-NMMA + L-arginine, ^b— $p < 0.05$ L-NMMA vs L-NMMA + L-arginine, ^c—control vs indomethacin, ⁺— $p < 0.05$ group II vs group IV or group V. EC_{50} values were interpolated from concentration-response curves shown in Figs. 1–5.

animals $46.17 \pm 4.6\%$ relaxation was observed at 10^{-5} M, was significantly more than hypercholesterolemic (group II) vessels ($36.25 \pm 3.10\%$).

Response to adenosine

Adenosine (10^{-13} – 10^{-4} M) produced a concentration-dependent biphasic response (Fig. 2A, Table 5) in all the groups of animals. In aortic-rings from animals on normal feed supplemented with herbs (group III), an augmentation of adenosine-induced relaxation was observed as compared to animals on control diet alone (group I). In the animals on cholesterol plus herbal supplemented (Group IV), a significant ($p < 0.05$) enhancement of relaxation was observed at higher

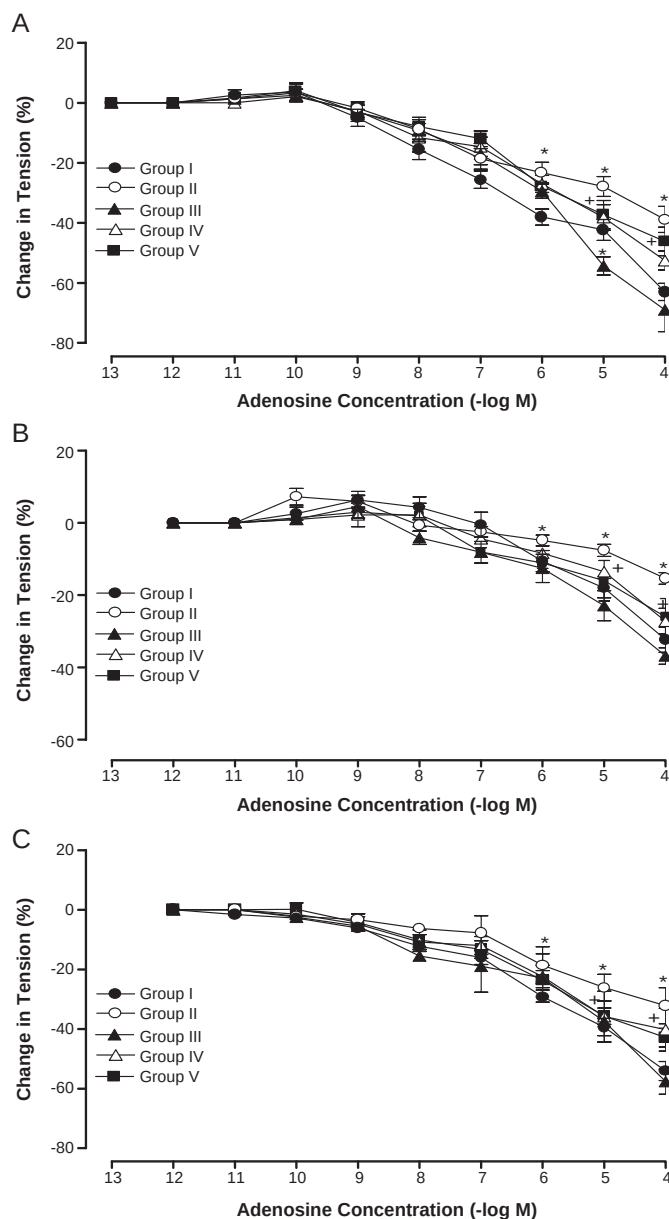


Fig. 2. Relaxation response to adenosine are expressed as decline (percent) in steady state tension obtained with 10^{-6} M PE. Concentration-response (10^{-13} – 10^{-4} M) to adenosine (A), after incubation with $30 \mu\text{M}$ L-NMMA (B) and 10^{-5} M L-arginine + $30 \mu\text{M}$ L-NMMA (C) in 10^{-6} M PE precontracted intact-endothelium aortic-rings from rats on normal feed (group I), cholesterol supplemented feed (group II), normal feed supplemented with turmeric + garlic (group III), cholesterol plus turmeric + garlic supplemented feed (group IV) and high cholesterol supplemented diet followed by turmeric + garlic plus cholesterol supplemented feed (group V). Data represents mean $\pm$ SEM, 4 rings from each animal. * $p < 0.05$ group I vs group II or group III, + group II vs group IV or group V.

Table 5

Effective concentration of adenosine for 50% response (EC_{50}) and maximum relaxant response—% change (E_{max}) to adenosine, before and after incubation of tissues with L-monomethylarginine (L-NMMA), L-NMMA + L-arginine, and indomethacin from rats on normal feed (group I), cholesterol supplemented feed (group II), normal feed supplemented with turmeric + garlic (group III), cholesterol plus turmeric + garlic supplemented feed (group IV) and high cholesterol supplemented diet followed by turmeric + garlic plus cholesterol supplemented feed (group V)

Groups	n	EC_{50} ($-\log M$)	E_{max}
<i>CONTROL</i>			
I	12	5.32 ± 0.28	80.3
II	10	—	48.7
III	10	6.82 ± 0.37	85.65
IV	8	8.62 ± 0.58	65.5
V	8	4.81 ± 0.63	62.33
<i>L-NMMA (30 μM)</i>			
I	12	5.85 ± 0.04^a	60.5
II	10	—	36.33
III	10	55.68 ± 0.07^a	65.5
IV	8	—	48.85
V	8	—	42.55
<i>L-NMMA (30 μM) + L-ARGININE (10^{-5} M)</i>			
I	12	$6.72 \pm 0.12^{a,b}$	86.2
II	10	$5.96 \pm 0.03^{*,a}$	53.23
III	10	$6.20 \pm 0.04^{*,a,b}$	92.5
IV	8	$5.38 \pm 0.11^{+,a}$	70.11
V	8	$0.35 \pm 0.09^{+,a}$	64.5
<i>INDOMETHACIN (10^{-5} M)</i>			
I	10	5.07 ± 0.04^c	77.2
II	8	5.88 ± 0.13	63.5
III	6	6.57 ± 0.07^c	82.5
IV	6	5.31 ± 0.07^c	68.5
V	5	5.37 ± 0.04^c	66.66

Data represent mean $\pm$ SEM, n = number of animals in each group, from each animal 4 rings were taken.

*— $p < 0.05$ group I vs group II or group III, ^a— $p < 0.05$ control vs L-NMMA or L-NMMA + L-arginine, ^b— $p < 0.05$ L-NMMA vs L-NMMA + L-arginine, ^c—control vs indomethacin, ⁺— $p < 0.05$ group II vs group IV or group V. EC_{50} values were interpolated from concentration-response curves shown in Figs. 1–5.

concentrations (10^{-6} – 10^{-4} M) as compared to hypercholesterolemic animals (group II). However, supplementation with herbs to the hypercholesterolemic animals (group V) resulted in a rightward shift in relaxation curve.

L-NMMA (30 μM) caused a leftward shift in the relaxation curve to adenosine in all the groups of animals (Fig. 2B, Table 5). However, the shift was small in groups of animals supplemented with herbs (group III, IV and V) as compared to hypercholesterolemic animals (group II). Aortic-rings from group IV animals showed a significant ($p < 0.05$) improvement in relaxation at higher concentrations (10^{-6} M and more) compared to group II animals. A significant ($p < 0.05$) 39.64 $\pm$ 4.55% relaxation was observed in group V animals at highest concentration (10^{-4} M) as compared to group II (28.62 $\pm$ 3.01%) animals.

Addition of L-arginine (10^{-5} M) to the tissues pretreated with L-NMMA (30 μ M) reversed the effects of NO blockade by L-NMMA (Fig. 2C, Table 5). The reversal of adenosine-induced relaxation was notable in group III animals. Response curve of animals on high cholesterol plus herbs supplemented (group IV and V) significantly shifted towards control (group I) after incubation with L-arginine. A maximum of $65.30 \pm 5.55\%$ relaxation was observed in group IV animals, $50.71 \pm 4.19\%$ in hypercholesterolemic (group II) animals.

Response to isoproterenol

Aortic ring from the animals on normal feed supplemented with herbs (group III) showed a marked relaxation of $76.49 \pm 2.17\%$ at highest concentration (10^{-4} M) of IP (Fig. 3A, Table 6). In aortic-rings from the control animals (group I), cholesterol plus herbal supplemented feed animals (group IV) and hypercholesterolemic animals on cholesterol plus herbal supplemented feed (group V), an augmentation of IP-induced relaxation was observed.

A significant ($p < 0.05$) attenuation of relaxation to IP was seen on incubation of aortic-rings with L-NMMA (30 μ M) in all the groups of animals (Fig. 3B). In animals from group I as well as group III, a significant ($p < 0.05$) increase in tension with PE (10^{-6} M) and increase in EC_{50} was observed (Table 6). In aortic-rings from group IV animals, the response-curve was similar to group II at lower concentrations (up to 10^{-8} M) while, higher concentrations showed more pronounced relaxation.

A reversal of attenuation of relaxation by L-NMMA was observed after incubation of tissues with L-arginine (10^{-5} M), in all the groups of animals (Fig. 3C, Table 6). In aortic-rings from group III animals, the reversal of IP-induced relaxation was prominent (100%) (Fig. 3C).

Response to 5-hydroxytryptamine

A concentration-dependent (10^{-13} – 10^{-4} M) contractile response was observed in aortic-rings precontracted with 10^{-6} M PE from all the groups of animals (Fig. 4A, Table 7). In aortic-rings from animals on standard diet with herbal supplement (group III), the contraction was small and was apparent only at higher concentrations (10^{-7} M and more). The response-curve to 5-HT in animals supplemented with cholesterol plus herbs (group IV) was similar to control animals (group I). In hypercholesterolemic animals, herbal supplement (group V) resulted in decrease in tension as compared to hypercholesterolemic (group II) animals (Fig. 4A).

L-NMMA (30 μ M) induced a significant leftward shift in concentration (10^{-13} – 10^{-4} M) response curves in all the groups of animals (Fig. 4B, Table 7). Aortic-rings from group III animals showed a response similar to the control (group I). However, in most of the rings contraction rose slowly and was apparent only at higher concentrations. Herbal supplementation (group IV and V) resulted in a significant attenuation of 5-HT-induced contraction at lower concentrations (up to 10^{-7} M) as compared to hypercholesterolemic (group II) animals.

Incubation of tissues with 10^{-5} M L-arginine partially reversed the effect of L-NMMA on response curve of 5-HT in all the groups of animals (Fig. 4C, Table 7). Aortic-rings from group III, IV and V animals produced, only 5% contraction up to 10^{-8} M concentration of 5-HT, which was significantly less than the values from hypercholesterolemic (group II) animals.

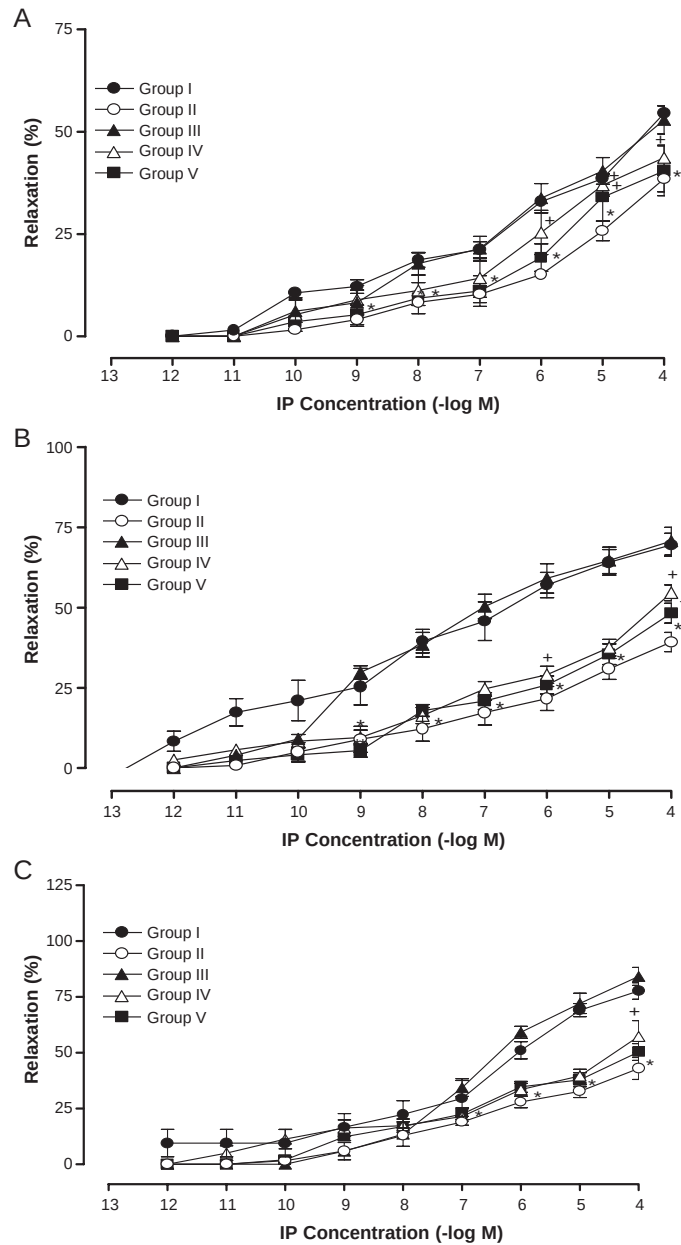


Fig. 3. Concentration-response (10^{-13} – 10^{-4} M) to isoproterenol (A), after incubation with 30 μ M L-NMMA (B) and 10^{-5} M L-arginine + 30 μ M L-NMMA (C) in 10^{-6} M PE precontracted intact-endothelium aortic-rings from rats on normal feed (group I), cholesterol supplemented feed (group II), normal feed supplemented with turmeric + garlic (group III), cholesterol plus turmeric + garlic supplemented feed (group IV) and high cholesterol supplemented diet followed by turmeric + garlic plus cholesterol supplemented feed (group V). Data represents mean $\pm$ SEM, 4 rings from each animal. * $p < 0.05$ group I vs group II or group III, + group II vs group IV or group V.

Table 6

Effective concentration of isoproterenol for 50% relaxation (EC_{50}) and maximum relaxant response—% change (E_{max}) to isoproterenol, before and after incubation of tissues with L-monomethylarginine (L-NMMA), L-NMMA + L-arginine, and indomethacin from rats on normal feed (group I), cholesterol supplemented feed (group II), normal feed supplemented with turmeric + garlic (group III), cholesterol plus turmeric + garlic supplemented feed (group IV) and high cholesterol supplemented diet followed by turmeric+garlic plus cholesterol supplemented feed (group V)

Groups	n	EC_{50} (– log M)	E_{max}
<i>CONTROL</i>			
I	12	8.38 ± 0.08	100
II	10	–	48.5
III	10	$9.97 \pm 0.07^*$	98.3
IV	8	6.87 ± 0.11	67.3
V	8	6.12 ± 0.13	56.5
<i>L-NMMA (30 μM)</i>			
I	12	6.72 ± 0.04^a	66.66
II	10	–	44.25
III	10	6.77 ± 0.09^a	71.5
IV	8	5.90 ± 0.12	60.2
V	8	4.83 ± 0.19^a	50.63
<i>L-NMMA (30 μM) + L-ARGININE (10^{-5} M)</i>			
I	12	8.95 ± 0.10	100.0
II	10	$4.73 \pm 0.13^*$	54.5
III	10	8.63 ± 0.04^b	100.0
IV	8	$6.59 \pm 0.19^{+a}$	85.5
V	8	$6.97 \pm 0.08^{+a}$	80.0
<i>INDOMETHACIN (10^{-5} M)</i>			
I	10	7.70 ± 0.09^c	90.0
II	8	$6.93 \pm 0.08^{*,c}$	70.0
III	6	$7.27 \pm 0.03^{*,c}$	85.25
IV	6	$6.28 \pm 0.06^{+,c}$	72.5
V	5	6.58 ± 0.11^c	69.3

Data represent mean $\pm$ SEM, n = number of animals in each group, from each animal 4 rings were taken.

*– $p < 0.05$ group I vs group II or group III, a – $p < 0.05$ control vs L-NMMA or L-NMMA + L-arginine, b – $p < 0.05$ L-NMMA vs L-NMMA + L-arginine, c –control vs indomethacin, $^+$ – $p < 0.05$ group II vs group IV or group V. EC_{50} values were interpolated from concentration-response curves shown in Figs. 1–5.

Effect of indomethacin on response to acetylcholine, adenosine, isoproterenol and 5-hydroxytryptamine

Incubation of tissues with (10^{-5} M) indomethacin, produced a significant leftward shift in response curves to ACh in all the groups (Fig. 5A; Table 4). A significant ($p < 0.05$) decrease in EC_{50} values was also observed in herbal supplemented animals (groups IV and V) as compared to hypercholesterolemic animals (group II) (Table 4). About 60% relaxation was recorded in group IV and V, which was found to be significantly more than the hypercholesterolemic animals (group II).

The addition of indomethacin (10^{-5} M) shifted the adenosine concentration-response curves to left in all the groups (Fig. 5B, Table 5). Aortic-rings from group III animals showed a significant ($p < 0.05$) increase in relaxation ($67.14 \pm 4.62\%$ at 10^{-5} M) as compared to control (group I) animals ($58.25 \pm$

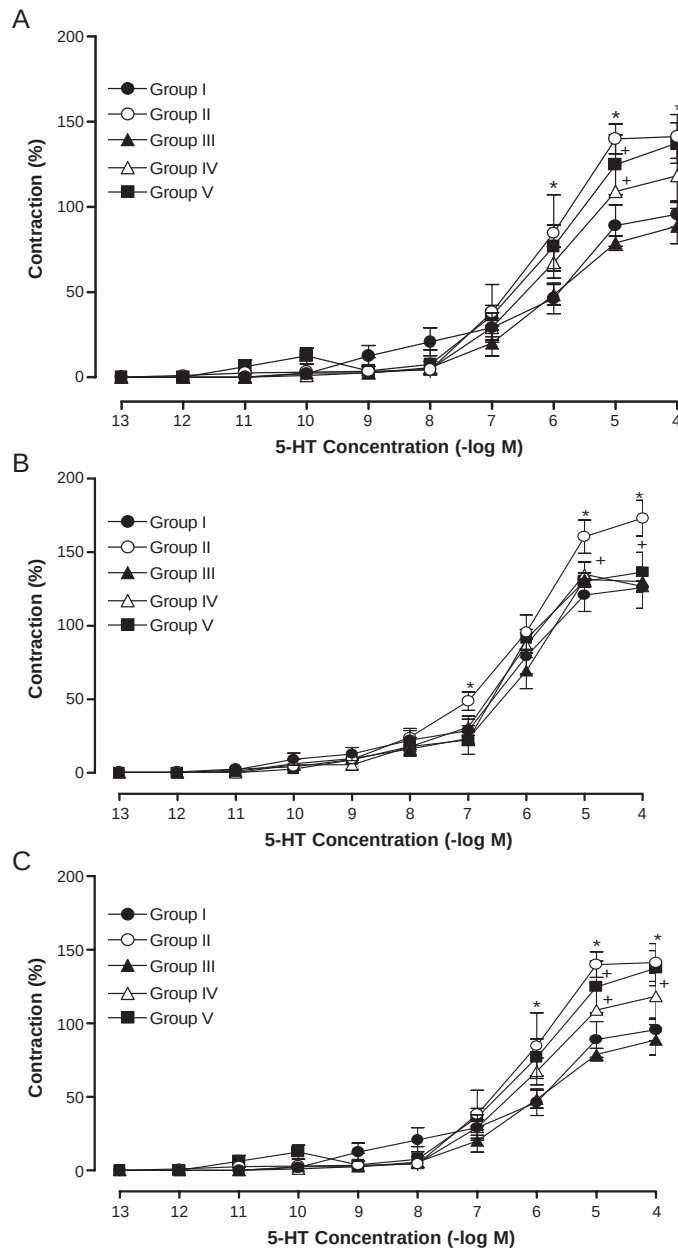


Fig. 4. The contractile-responses to 5-hydroxytryptamine (5-HT) are expressed as increase in (percent) steady state tension obtained with 10^{-6} M PE. Concentration-response (10^{-13} – 10^{-4} M), to 5-HT (A), after incubation with 30 μ M L-NMMA (B) and 10^{-5} M L-arginine + 30 μ M L-NMMA (C) in 10^{-6} M PE precontracted intact-endothelium aortic-rings from rats on normal feed (group I), cholesterol supplemented feed (group II), normal feed supplemented with turmeric + garlic (group III), cholesterol plus turmeric+garlic supplemented feed (group IV) and high cholesterol supplemented diet followed by turmeric + garlic plus cholesterol supplemented feed (group V). Data represents mean $\pm$ SEM, 4 rings from each animal. * $p < 0.05$ group I vs group II or group III, + group II vs group IV or group V.

Table 7

Effective concentration of isoproterenol for 50% relaxation (EC_{50}) and maximum contractile response-% change (E_{max}) to 5-hydroxytryptamine, before and after incubation of tissues with L-monomethylarginine (L-NMMA), L-NMMA + L-arginine, and indomethacin from rats on normal feed (group I), cholesterol supplemented feed (group II), normal feed supplemented with turmeric + garlic (group III), cholesterol plus turmeric + garlic supplemented feed (group IV) and high cholesterol supplemented diet followed by turmeric + garlic plus cholesterol supplemented feed (group V)

Groups	n	EC_{50} ($-\log M$)	E_{max}
<i>CONTROL</i>			
I	12	7.57 ± 0.1	112.5
II	10	$8.73 \pm 0.06^*$	152.5
III	10	7.44 ± 0.12	130.0
IV	8	$7.02 \pm 0.09^+$	138.3
V	8	$7.18 \pm 0.13^+$	147.77
<i>L-NMMA (30 μM)</i>			
I	12	7.42 ± 0.04	143.5
II	10	$7.02 \pm 0.13^{*,a}$	188.8
III	10	7.58 ± 0.09	150.0
IV	8	7.23 ± 0.08	170.0
V	8	7.27 ± 0.07	178.25
<i>L-NMMA (30 μM) + L-ARGININE (10^{-5} M)</i>			
I	12	6.08 ± 0.04^a	108.0
II	10	$7.25 \pm 0.06^{*,b}$	138.5
III	10	$7.05 \pm 0.09^{*,a}$	110.0
IV	8	7.13 ± 0.13	125.46
V	8	7.21 ± 0.08	135.5
<i>INDOMETHACIN (10^{-5} M)</i>			
I	10	6.64 ± 0.06^c	93.5
II	8	$6.04 \pm 0.05^{*,c}$	115.55
III	6	$6.76 \pm 0.13^{*,c}$	79.25
IV	6	$6.47 \pm 0.06^{+,c}$	102.5
V	5	6.18 ± 0.08^c	108.3

Data represent mean $\pm$ SEM, n = number of animals in each group, from each animal 4 rings were taken.

*-p < 0.05 group I vs group II or group III, ^a-p < 0.05 control vs L-NMMA or L-NMMA + L-arginine, ^b-p < 0.05 L-NMMA vs L-NMMA + L-arginine, ^c-control vs indomethacin, ⁺-p < 0.05 group II vs group IV or group V. EC_{50} values were interpolated from concentration-response curves shown in Figs. 1–5.

1.99%). Tissues from group IV and V showed a significant ($p < 0.05$) increase in relaxation as compared to hypercholesterolemic (group II) animals.

Addition of indomethacin (10^{-5} M) augmented the IP-induced relaxation in all the groups of animals (Fig. 5C, Table 6). Rings from group III animals showed a marked relaxation (E_{max} 85.25) and in some tissues (n = 4), maximum relaxation was recorded only at 10^{-7} M concentration. A significantly ($p < 0.05$) enhanced relaxation was observed in group IV and V animals as compared to hypercholesterolemic (group II) animals. Even 10^{-8} M concentration produced, about 30% relaxation vs $11.66 \pm 0.72\%$ in hypercholesterolemic animals. The EC_{50} values significantly ($p < 0.05$) decreased in group IV and V as compared to the group II animals ($p < 0.05$) (Table 6).

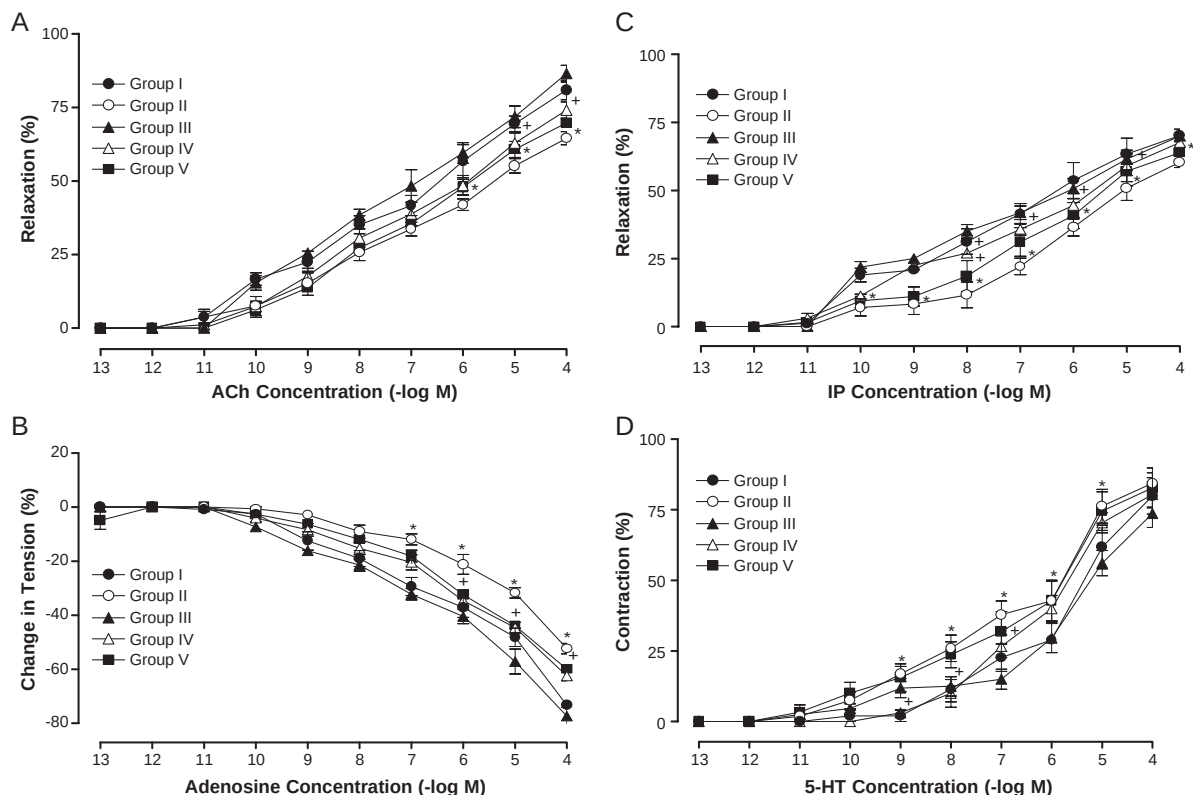


Fig. 5. Concentration-response (10^{-13} – 10^{-4} M) to acetylcholine (A), adenosine (B), isoproterenol (C) and 5-HT (D) after incubation with 10^{-5} M indomethacin, in 10^{-6} M PE precontracted intact-endothelium aortic-rings, from rats on normal feed (group I), cholesterol supplemented feed (group II), normal feed supplemented with turmeric + garlic (group III), cholesterol plus turmeric + garlic supplemented feed (group IV) and high cholesterol supplemented diet followed by turmeric + garlic plus cholesterol supplemented feed (group V). Data represents mean $\pm$ SEM, 4 rings from each animal. * $p < 0.05$ group I vs group II or group III, + group II vs group IV or group V.

Incubation of tissues with 10^{-5} M indomethacin shifted the concentration-response curves of 5-HT to right in all the groups of animals (Fig. 5D; Table 7). In aortic-rings from group III animals, a decrease in contractile response was observed and the response was closer to group I animals. Contractile response to 5-HT was significantly smaller in group IV and V as compared to group II animals. In group IV animals the 5-HT-induced contraction was small at lower concentrations (up to 10^{-9} M).

Discussion

Natural compounds have been proved to be useful in the treatment of different human pathological conditions. The scientific consideration of therapeutic potential of these natural herbs is still inappropriate due to lack of proper understanding of mechanism of their action. Further knowledge of pathophysiology of impaired endothelium and the prophylactic and therapeutic properties of natural compounds will contribute to reveal the remaining mystery of many cardiovascular diseases.

Blood pressure and blood lipid contents probably are the major determinants of the tendency of lipid deposition in the arterial intima. Hence an attempt was made to assess the cardiovascular response in animals fed with cholesterol diet and herbal supplemented diets. Although there is no direct relationship between the degree of hypertension and the extent of atherosclerotic lesions (James, 1974), there are evidences that hypertension per se is responsible for the extent of atheroma formation (James, 1974; Genest et al., 1978). Baroreceptor resetting due to atherosclerosis might result in increased mean blood pressure observed by us (Zimmerman and Frohlich, 1990).

A notable restoration of arterial blood pressure was seen in animals on herbal supplemented diet. The hypotensive effect of garlic and turmeric supported the earlier studies (Ramirez-Tortosa et al., 1999) suggests the beneficial role of these herbs in cardiovascular disorders. The exact mechanism responsible for lipid reducing property is not known. However, the synergistic effect of both antihypertensive constituents (Orekhov et al., 1995; Azuine and Bhide, 1992), may be responsible for this. The electrocardiogram (ECG) did not show any significant change in the pattern after cholesterol supplemented feed and after treatment with garlic and turmeric. Only mild tachycardia was observed in hypercholesterolemic animals. This does not point out any cardiac abnormality.

For last few decades, herbal drugs have been used by patients as a cholesterol reducing agents. Elevated levels of serum total cholesterol and serum LDL are the primary risk factors for atherosclerosis and hypertension.

Our study demonstrated an eight fold increases in serum cholesterol levels in hypercholesterolemic animals. A corresponding increase in serum triglycerides, LDL and VLDL levels occurred in hypercholesterolemic animals and serum HDL level fell drastically. This is in agreement with earlier observation by Chen et al. (2002).

Garlic and turmeric are known to retard the development of atherosclerosis (Raj kumar et al., 1998). Our study demonstrated that these herbs might exert its beneficial effect partially by reducing the atherogenic properties of cholesterol and maintaining the NO-mediated endothelial function possibly by inhibiting the oxidation of LDL and by increasing serum HDL. The results documented in Table 2 strongly indicate that the herbal supplement decreased the triglyceride levels even in hypercholesterolemic animals, which may lead to suppression of atherogenesis (Bradley et al., 1994).

Garlic is reported to decrease serum cholesterol, LDL and TG, possibly through inhibition of HMG-CoA reductase activity (Orekhov et al., 1995). Curcumin increases the excretion of cholesterol and bile acids (Raj kumar et al., 1998; Rao et al., 1970).

Thus, our observations and earlier studies suggest that the rise in atherogenic index (total cholesterol/HDL) due to high cholesterol diet enhances the probability of cardiovascular pathogenesis and endothelial dysfunction, which can be partially managed by treatment with natural herbs. There was no significant change in body weight and daily feed intake of the animals, suggesting the normal metabolic state of the animals. The vasorelaxant response of turmeric appears to be mediated through endothelium, supporting an anti-oxidant activity as well as a sharp transient hypotensive effect of turmeric (Azuine and Bhide, 1992).

Animals on supplemented feed showed a significantly enhanced vasorelaxant response to adenosine, ACh and IP and contractile effect of 5-HT was significantly attenuated, indicating a prophylactic action of herbs in protecting/restoring the normal function of endothelium. Earlier studies have suggested that garlic enhances NOS activity (Das et al., 1995). This view is further strengthened by the observation that the attenuation of responses to IP, ACh and adenosine by L-NMMA was smaller in tissues from herbal treated animals. This is in agreement with earlier studies by Chaveri and coworker (Pedraza-Chaverri et

al., 1998) and the report that the therapeutic action of garlic is mainly through NO (Das et al., 1995). Turmeric is also reported to inhibit the nitrite production from NO in mouse peritoneum, suggesting the role of turmeric in maintaining vasorelaxant response. The pronounced response to adenosine observed, could be due to the action of endogenous adenosine present in garlic (Agarwal, 1996). However, Das et al., reported that neither L-arginine nor alliin derived allyl compounds are responsible for activation of NOS in cell free system (Das et al., 1996).

The increase in relaxant response to ACh, adenosine and IP after incubation of tissues with a cyclooxygenase blocker, indomethacin was prominent in tissues from garlic and turmeric treated animals. These results support the earlier study (Das et al., 1995) that garlic inhibits cyclooxygenase activity and it has been reported that indomethacin enhances the antiplatelet activity of ajoene; one of the active molecules of garlic (Agarwal, 1996). Turmeric is also known to inhibit platelet aggregation (Chan et al., 1995) thereby, inhibiting the contractile factors released due to endothelial damage in atherosclerosis.

Finally, indigenous drugs are the natural source of protection against certain disorders, which can be used more effectively by the knowledge of their active ingredients as well as by their mechanism of action. The present study demonstrated that garlic and turmeric are potent vasorelaxants as well as reduce the atherogenic properties of cholesterol. Whether combination of these vasodilator natural herbs is useful in treatment of cardiovascular disorders with increased peripheral vascular resistance remains to be determined.

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References

- Agarwal, K.C., 1996. Therapeutic actions of garlic constituents. *Medical Research Reviews* 16 (1), 111–124.
- Alder, R., Lookinland, S., Berry, J.A., Williams, M., 2003. A systematic review of the effectiveness of garlic as an anti-hyperlipidemic agent. *Journal of American Academy of Nurse Practitioners* 15 (3), 120–129.
- Antoniades, C., Tousoulis, D., Tentolouris, C., Toutouzias, P., Stefanadis, C., 2003. Oxidative stress, antioxidant vitamins, and atherosclerosis: From basic research to clinical practice. *Herz* 28 (7), 628–638.
- Aqel, M.B., Gharaibah, M.N., Salhab, A.S., 1991. Direct relaxant effects of garlic juice on smooth and cardiac muscles. *Journal of Ethnopharmacology* 33 (1–2), 13–19.
- Ashraf, M.Z., Khan, M.S., Hameed, H.A., Hussain, M.E., Fahim, M., 1999. Endothelium modulated vasorelaxant response of a polypharmaceutical herbal drug (lipotab) and its individual constituents. *Journal of Ethnopharmacology* 66 (1), 97–102.
- Azuine, M.A., Bhide, S.V., 1992. Chemopreventive effect of turmeric against stomach and skin tumors induced by chemical carcinogens in swiss mice. *Nutrition Cancer* 17 (1), 77–83.
- Banerjee, S.K., Maulik, S.K., 2002. Effect of garlic on cardiovascular disorders: A review. *Nutrition Journal* 1 (1), 4.
- Biegelsen, E.S., Loscalzo, J., 1999. Endothelial function and atherosclerosis. *Coronary Artery Diseases* 10 (4), 241–256.
- Brace, L.D., 2002. Cardiovascular benefits of garlic (*Allium sativum* L.). *Journal of Cardiovascular Nursing* 16 (4), 33–49.

- Bradley, W.A., Booyse, F.M., Gianturco, S.H., 1994. Fibrinolytic and thrombotic factors in atherosclerosis and IHD: The influence of triglyceride rich lipoproteins (TGRLP). *Atherosclerosis* 108 (Suppl.), S31–S39.
- Chan, M.M., Ho, C.T., Huang, H.I., 1995. Effects of three dietary phytochemicals from tea, rosemary and turmeric on inflammation-induced nitrite production. *Cancer Letters* 96 (1), 23–29.
- Chen, M., Masaki, T., Sawamura, T., 2002. Lox-1, the receptor for oxidized low-density lipoprotein identified from endothelial cells: Implications in endothelial dysfunction and atherosclerosis. *Pharmacology and Therapeutics* 95 (1), 89–100.
- Das, I., Hirani, J., Sooranna, S., 1996. Arginine is not responsible for the activation of nitric oxide synthase by garlic. *Journal of Ethnopharmacology* 53 (1), 5–9.
- Das, I., Khan, N.S., Sooranna, S.R., 1995. Potent activation of nitric oxide synthase by garlic: A basis for its therapeutic applications. *Current Medical Research and Opinion* 13 (5), 257–263.
- Deters, M., Klabunde, T., Meyer, H., Resch, K., Kaever, V., 2003. Effects of curcumin on cyclosporine-induced cholestasis and hypercholesterolemia and on cyclosporine metabolism in the rat. *Planta Medica* 69 (4), 337–343.
- Elkayam, A., Mirelman, D., Peleg, E., Wilchek, M., Miron, T., Rabinkov, A., Oron-Herman, M., Rosenthal, T., 2003. The effects of allicin on weight in fructose-induced hyperinsulinemic, hyperlipidemic, hypertensive rats. *American Journal of Hypertension* 16 (12), 1053–1056.
- Fahim, M., Arndt, J.O., 1990. Left atrial receptors in arterial baroreflex control of heart rate. *Japanese Journal of Physiology* 40 (1), 33–55.
- Fahim, M., Hussain, T., Mustafa, S.J., 2001. Relaxation of rat aorta by adenosine in diabetes with and without hypertension: Role of endothelium. *European Journal of Pharmacology* 412 (1), 51–59.
- Fan, J., Watanabe, T., 2003. Inflammatory reactions in the pathogenesis of atherosclerosis. *Journal of Atherosclerosis and Thrombosis* 10 (2), 63–71.
- Genest, J., Nowaczynski, W., Boucher, R., Kuchel, O., 1978. Role of the adrenal cortex and sodium in the pathogenesis of human hypertension. *Canadian Medical Association Journal* 118 (5), 538–549.
- Ghasi, S., Nwobodo, E., Ofili, J.O., 2000. Hypocholesterolemic effects of crude extract of leaf of moringa oleifera lam in high-fat diet fed wistar rats. *Journal of Ethnopharmacology* 69 (1), 21–25.
- Harenberg, J., Giese, C., Zimmermann, R., 1988. Effect of dried garlic on blood coagulation, fibrinolysis, platelet aggregation and serum cholesterol levels in patients with hyperlipoproteinemia. *Atherosclerosis* 74 (3), 247–249.
- James, T.N., 1974. Diseases of the large and small coronary arteries. *Archives of Internal Medicine* 134 (1), 163–176.
- Kitamoto, S., Egashira, K., 2004. Endothelial dysfunction and coronary atherosclerosis. *Current Drug Targets: Cardiovascular and Haematological Disorders* 4 (1), 13–22.
- Kris-Etherton, P.M., Etherton, T.D., Carlson, J., Gardner, C., 2002. Recent discoveries in inclusive food-based approaches and dietary patterns for reduction in risk for cardiovascular disease. *Current Opinion in Lipidology* 13 (4), 397–407.
- Landmesser, U., Hornig, B., Drexler, H., 2004. Endothelial function: A critical determinant in atherosclerosis? *Circulation* 109 (21 Suppl. 1), II27–II33.
- Maslin, D.J., Brown, C.A., Das, I., Zhang, X.H., 1997. Nitric oxide—a mediator of the effects of garlic? *Biochemical Society Transactions* 25 (3), 408S.
- Orekhov, A.N., Tertov, V.V., Sobenin, I.A., Pivovarova, E.M., 1995. Direct anti-atherosclerosis-related effects of garlic. *Annals of Medicine* 27 (1), 63–65.
- Pedraza-Chaverri, J., Tapia, E., Medina-Campos, O.N., de los Angeles Granados, M., Franco, M., 1998. Garlic prevents hypertension induced by chronic inhibition of nitric oxide synthesis. *Life Sciences* 62 (6), PL71–PL77.
- Raj kumar, S., Pillai, K.K., Balani, D.K., Hussain, S.Z., 1998. Antiatherosclerotic effect of lipotab forte in cholesterol-fed rabbits. *Journal of Ethnopharmacology* 59 (3), 125–130.
- Ramirez-Tortosa, M.C., Mesa, M.D., Aguilera, M.C., Quiles, J.L., Baro, L., Ramirez-Tortosa, C.L., Martinez-Victoria, E., Gil, A., 1999. Oral administration of a turmeric extract inhibits LDL oxidation and has hypocholesterolemic effects in rabbits with experimental atherosclerosis. *Atherosclerosis* 147 (2), 371–378.
- Rao, D.S., Sekhara, N.C., Satyanarayana, M.N., Srinivasan, M., 1970. Effect of curcumin on serum and liver cholesterol levels in the rat. *Journal of Nutrition* 100 (11), 1307–1315.
- Rosenson, R.S., 2001. Pluripotential mechanisms of cardioprotection with HMG-CoA reductase inhibitor therapy. *American Journal of Cardiovascular Drugs* 1 (6), 411–420.
- Rubanyi, G.M., 1993. The role of endothelium in cardiovascular homeostasis and diseases. *Journal of Cardiovascular Pharmacology* 22 (Suppl. 4), S1–S14.

- Stokes, K.Y., Cooper, D., Tailor, A., Granger, D.N., 2002. Hypercholesterolemia promotes inflammation and microvascular dysfunction: Role of nitric oxide and superoxide. *Free Radical Biology and Medicine* 33 (8), 1026–1036.
- Thompson Coon, J.S., Ernst, E., 2003. Herbs for serum cholesterol reduction: A systematic view. *Journal of Family Practice* 52 (6), 468–478.
- Vanhoutte, P.M., Boulanger, C.M., 1995. Endothelium-dependent responses in hypertension. *Hypertension Research* 18 (2), 87–98.
- Zimmerman, R.S., Frohlich, E.D., 1990. Stress and hypertension. *Journal of Hypertension Suppl.* 8 (4), S103–S107.